Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/04/2021
PI Name:	Katabi, Dina Ph.D.		
Project Title:	Non-Contact Sleep, Vitals, and Behavior Sensing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02142	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	05/01/2019	**End Date:**	09/30/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2020 (original end date was 4/20/2020) per TRISH (Ed., 5/28/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0011		
Performance Goal No.:			
Performance Goal Text:			
	Industry Project We propose a new solution for noncontact health monitoring. The Emerald device is a novel technology that uses wireless signals to continuously monitor a variety of health metrics without requiring subjects to wear sensors or wearables. It is a Wi-Fi like device that sits in the background and analyzes the surrounding radio signals with advanced machine learning algorithms. It then extracts from these wireless signals a variety of metrics related to gait, mobility, behavior, breathing, sleep (including sleep stages and sleep apnea), scratching, etc. Over the past several years, our team has developed the research underlying the Emerald system, and validated its capabilities to accurately extract all of these physiological metrics. We have also collaborated with doctors and clinical researchers to demonstrate the applicability of these measured metrics to actual patient health outcomes in a variety of therapeutic areas. The current system has been targeted towards running short experiments in a lab, clinic, or controlled environment. This proposal focuses on taking		

Task Description:	the next step -- enabling the Emerald system to be used for continuous and interactive health monitoring in the person's natural living environment. Doing this requires operationalization of the technology along three axes: - First, the system should run continuously without requiring regular manual intervention from engineers. This requires building a software pipeline that automatically manages data collection, cleaning, analysis, and storage. Further, this pipeline needs to be robust to deal with any intermittent failures or glitches in the processing. - Second, the system needs to be able to create a per-user profile that maintains the history of the user's health metrics over time, analyzes it for changes from baseline, and reports any deviations from normal. - Finally, the system should allow the user to access his/her health status in an easy and friendly manner that digests the analysis into understandable, meaningful, and actionable results. This requires designing and implementing a user interface that shows the user both their health metrics and deviations from their baselines, and also allows them to dive in and zoom into various aspects of their health metrics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Emerald is a novel technology that allows for monitoring people's health and physiological signals in a passive way without asking them to wear sensors on their body or measure themselves. The technology is based on a wireless device that looks like WiFi router. It transmits very low power radio signals, which bounce off people and objects in the environment. The device analyzes these reflected radio signals using machine learning algorithms and infers people's sleep, respiration, and movements. The Emerald health metrics are validated in comparison to Federal Aviation Administration (FAA) approved methods for measuring respiration, sleep, and motion. Our team has developed the technology at Massachusetts Institute of Technology (MIT) and created a spin out to commercialize it for improving health and wellness. The Emerald technology is well-suited for a physical health monitoring system measuring indicators of health and human performance of astronauts on long duration deep space missions. Specifically, NASA has Sleep loss and circadian desynchronization as key risks to space mission that affect astronauts' performance and lead to adverse health outcomes. It has also highlighted the need to identify a set of validated and minimally obtrusive tools to monitor and measure sleep-wake activity and associated performance changes for spaceflight. The Emerald technology promises to address this risk and deliver the unmet need. Specifically, it delivers a simple contactless sensor that monitors sleep-awake activities, and reports sleep stages, sleep efficiency, and changes in the circadian cycle. The sensor is unobtrusive and does not require astronauts to wear sensors or actively report their sleep. They can go about their mission while the Emerald sensor passively monitor their sleep. This project focuses on technical development that allows the use of Emerald technology for monitoring sleep during space missions. While the key algorithms have been proven to accurately infer sleep stages and other sleep parameters, prior to this project, the technology was not ready for real-world deployment. Specifically, past research required a significant amount of manual processing where our engineers and data scientists manually collect the data, reformat the collected data, run the algorithms, schedule different processes, and manage computation and storage processes. Thus, in this project we developed the key infrastructure to automate all of these processes and transform the Emerald technology from a system capable of only short term, non-interactive studies to a platform that can be used for continuous, long-term, longitudinal sleep monitoring. To this end, we have developed the following three components: 1) an automated pipeline for data collection and processing, 2) a storage system and access mechanisms that operate on per-user sleep analytics, 3) and a secure user interface to allow users to access their data. The project was also extended to address new needs that arose from the COVID pandemic. Emerald's ability to monitor physiological signals, e.g., respiration, became particularly relevant for monitoring COVID-19 patients from a distance without any body contact, and hence without incurring contagion risks. Thus, the project was extended to develop tools to track respiration and provide an alert system that allows a medical doctor to configure the system to remotely monitor COVID patients, and to send a notification when certain conditions of abnormal breathing are detected. The extension also covers an observational study in which the Emerald sensor is used to monitor actual COVID patients to check the viability of using Emerald for passive monitoring of COVID patients and tracking their recovery process and identifying recovery problems.
Task Progress:	We propose a new solution for noncontact health monitoring. The Emerald device is a novel technology that uses wireless signals to continuously monitor a variety of health metrics without requiring subjects to wear sensors or wearables. It is a Wi-Fi like device that sits in the background and analyzes the surrounding radio signals with advanced machine learning algorithms. It then extracts from these wireless signals a variety of metrics related to gait, mobility, behavior, breathing, sleep (including sleep stages and sleep apnea), scratching, etc. Over the past several years, our team has developed the research underlying the Emerald system, and validated its capabilities to accurately extract all of these physiological metrics. We have also collaborated with doctors and clinical researchers to demonstrate the applicability of these measured metrics to actual patient health outcomes in a variety of therapeutic areas. The current system has been targeted towards running short experiments in a lab, clinic or controlled environment. The objective of this project is to develop key components necessary for using this technology to monitor sleep during space missions. To do so, the project addresses the following three tasks: 1) automating data collection and processing, 2) developing storage and access of per user sleep analytics, 3) developing a secure user interface for accessing the data. A secondary goal of this project is to demonstrate the viability of using this technology to monitor COVID-19 patients in a contactless manner without any risk of contagion. This is done by using the technology to monitor 20 COVID-19 patients and control subjects, and developing a supportive alert system that tracks patient respiration and notifies the caregiver of alarming changes. This document provides a final report of the project, summarizing the progress made on the above primary and secondary goals. Specifically, the project has been executed successfully; all promised system development has been completed and tested. Additionally, the COVID respiration alert system has been developed and the COVID study has been conducted and demonstrated the importance of respiration monitoring.
Bibliography Type:	Description: (Last Updated: 01/13/2021)

Significant Media Coverage	Rosen A. "Tech from MIT may allow caregivers to monitor coronavirus patients from a distance. Article reports on the use of Emerald (PI's company) in monitoring COVID patients in retirement homes." Boston Globe, April 23, 2020. https://www.bostonglobe.com/2020/04/23/nation/tech-out-mit-may-allow-caregivers-monitor-coronavirus-patients-distance , Apr-2020